FIITJEE Students From All Programs have bagged 34 in Top 100, 66 in Top 200 and 174 in Top 500 All India Ranks. FIITJEE Performance in JEE (Advanced), 2014:
2521 FIITJEE Students from Classroom / Integrated School Programs & 3579 FIITJEE Students from All Programs have qualified in JEE (Advanced), 2014.

ALL INDIA INTEGRATED TEST SERIES

FIITJEE

JEE (Advanced), 2015

HALF COURSE TEST –VII

Paper 2

Time Allotted: 3 Hours **Maximum Marks: 243**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part is further divided into two sections: **Section-A & Section-B**
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

(i) **Section-A (01 to 09)** contains 09 multiple choice questions which have only one correct answer. Each question carries **+3 marks** for correct answer and **– 1 mark** for wrong answer.

Section-A (10 – 13) contains 4 Assertion-Reasoning (multiple choice questions) which have only one correct answer. Each question carries **+3 marks** for correct answer and **– 1 mark** for wrong answer.

Section-A (14 – 19) contains 2 paragraphs. Based upon paragraph, 3 multiple choice questions have to be answered. Each question has only one correct answer and carries **+4 marks** for correct answer and **– 1 mark** for wrong answer.

(ii) **Section-B (1 – 03)** contains 3 Matrix Match Type (4 × 4 Matrix) questions containing statements given in 2 columns. Statements in the first column have to be matched with statements in the second column. Each question carries **+6 marks** for all correct answer. There is no negative marking.

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

Physics

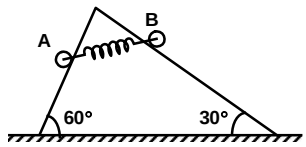
PART – I

SECTION – A

Straight Objective Type

This section contains **9 multiple choice questions** numbered 1 to 9. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct

1. Two uniform solid cylinders A and B each of mass 1 kg are connected by a light spring of force constant 200 N/m at their axles and are placed on a fixed wedge as shown in the figure. The coefficient of friction between the wedge and the cylinders is 0.2. The angle made by the line AB with the horizontal, in equilibrium, is

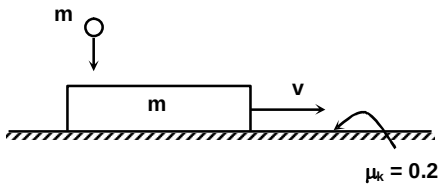


(A) 0° (B) 15°
(C) 30° (D) None of these.

2. A person wants to drive on the vertical surface of a large cylindrical wooden 'well' commonly known as 'deathwell' in a circus. The radius of the 'well' is 2 meter, and the coefficient of friction between the tyres of the motorcycle and the wall of the well is 0.2. Minimum speed the motorcyclist must have in order to prevent slipping should be (take g = 10 m/s^2)

(A) 10 m/s (B) 15 m/s
(C) 20 m/s (D) 25 m/s

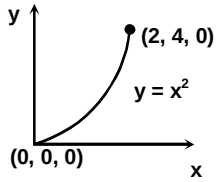
3. A ball of mass m falls vertically from a height h and collides with a block of equal mass m moving horizontally with a velocity v on a surface. The coefficient of kinetic friction between the block and the surface is $\mu_k = 0.2$, while the coefficient of restitution (e) between the ball and the block is 0.5. There is no friction acting between the ball and the block. The velocity of the block just after the collision decreases by



(A) $0.5\sqrt{2gh}$ (B) 0
(C) $0.1\sqrt{2gh}$ (D) $0.3\sqrt{2gh}$

Space for rough work

4. By applying a force $\vec{F} = (3xy - 5z)\hat{j} + 4z\hat{k}$ a particle is moved along the path $y = x^2$ from point (0, 0, 0) to the point (2, 4, 0). The work done by the F on the particle is



(A) $\frac{280}{5}$ (B) $\frac{140}{5}$

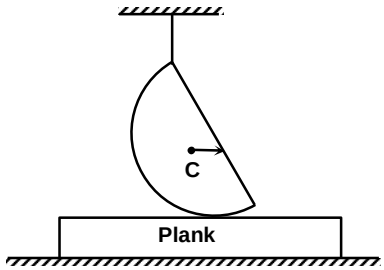
(C) $\frac{232}{5}$ (D) $\frac{192}{5}$

5. A bowler throws a ball horizontally along east direction with speed of 144 km/hr. The batsman hits the ball such that it deviates from its initial direction of motion by 74º north of east direction, without changing its speed. If mass of the ball is $\frac{1}{3}$ kg and time of contact between bat and ball is 0.02 s. Average force applied by batsman on ball is

(A) 800 N, 53º East of North (B) 800 N, 53º North of East

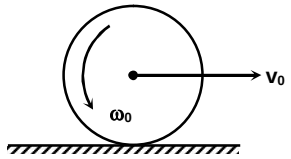
(C) 800 N, 53º North of West (D) 800 N, 53º West of North.

6. Lower surface of a plank is rough and lying at rest on a rough horizontal surface. Upper surface of the plank is smooth and has a smooth hemisphere placed over it through a light string as shown in the figure. After the string is burnt, trajectory of C.M. of the sphere is



(A) a circle
(B) a ellipse
(C) a straight line
(D) a parabola

7. A disc of radius R is spun to an angular speed ω_0 about its axis and then imparted a horizontal velocity of magnitude $\omega_0 R/4$ (at t = 0) with its plane remaining vertical. The coefficient of friction between the disc and the plane is μ. The sense of rotation and direction of its linear speed are shown in the figure. Choose the correct statement. The disc will return to its initial position



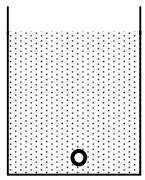
(A) if the value of $\mu < 0.5$. (B) irrespective of the value of μ ($\mu > 0$).

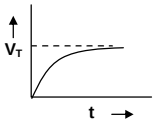
(C) if the value of $0.5 < \mu < 1$. (D) if $\mu > 1$.

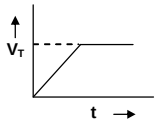
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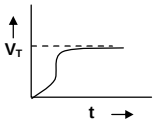
8. A particle is projected from the horizontal surface in earth's gravitational field at an angle θ with the horizontal then
   (A) center of curvature of projectile's trajectory at the highest point is below the horizontal surface level if $\theta < \tan^{-1}\sqrt{2}$
   (B) center of curvature of projectile's trajectory at the highest point is above horizontal surface if $\theta < \tan^{-1} 2$
   (C) center of curvature of projectile's trajectory at the highest point is below the ground level is $\theta > \tan^{-1} 2$
   (D) irrespective of the value of θ centre of curvature always lies below the horizontal surface of the earth.

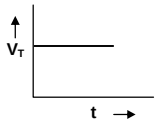
9. A small particle is released inside the depth of a liquid in the vessel. The volume of the particle is V. Density and viscosity of the liquid are ρ and η respectively. Buoyant force on the particle is greater than the weight of the particle. The variation of velocity of the particle with time is



(A) 

(B) 

(C) 

(D) 

Space for rough work

Reasoning Type

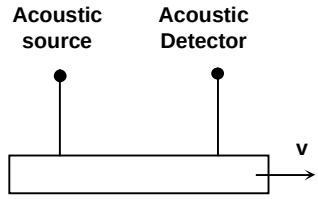
This section contains **4 questions** numbered 10 to 13. Each question contains STATEMENT-1 (Assertion) and STATEMENT-2 (Reason). Each question has 4 choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct

10. STATEMENT-1: In a standing wave formed in a stretched wire, the energy of each element of the wire remains constant.
and
STATEMENT-2: The net energy transfer in a standing wave is zero.
(A) Statement-1 is True, Statement-2 is True; Statement -2 **is** a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement -2 **is NOT** a correct explanation for Statement-1.
(C) Statement -1 is True, Statement-2 is False.
(D) Statement -1 is False, Statement-2 is True.

11. STATEMENT-1: The molecules on the surface of liquid filled in a container always experience a force normal to the surface if viscosity is absent.
and
STATEMENT-2: The liquid cannot sustain tangential strain.
(A) Statement-1 is True, Statement-2 is True; Statement -2 **is** a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement -2 **is NOT** a correct explanation for Statement-1.
(C) Statement -1 is True, Statement-2 is False.
(D) Statement -1 is False, Statement-2 is True.

Space for rough work

12. STATEMENT-1: If a plank is moving with speed v (v is greater than speed of sound). Acoustic source and detector are fixed as shown. Acoustic detector will never detect the sound emitted by acoustic source.



and

STATEMENT-2: The speed of sound is reference frame dependent.

(A) Statement-1 is True, Statement-2 is True; Statement -2 **is** a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement -2 **is NOT** a correct explanation for Statement-1.
(C) Statement -1 is True, Statement-2 is False.
(D) Statement -1 is False, Statement-2 is True.

13. STATEMENT-1: When a displacement wave in a string is reflected back from a rigid end, its phase is changed by π.

and

STATEMENT-2: At rigid end, impulsive elastic force changes its phase.

(A) Statement-1 is True, Statement-2 is True; Statement -2 **is** a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement -2 **is NOT** a correct explanation for Statement-1.
(C) Statement -1 is True, Statement-2 is False.
(D) Statement -1 is False, Statement-2 is True.

Space for rough work

Comprehension Type

This section contains **2 groups of questions.** Each group has 3 multiple choice question based on a paragraph. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** is correct

$P_{14–16}$: Paragraph for Question Nos. 14 to 16

When a particle is projected at some angle with the horizontal, the path of the particle is parabolic. In the process the horizontal velocity remains constant but the magnitude of vertical velocity changes. At any instant during flight the acceleration of the particle remains g in vertically downward direction. During flight at any point the path of particle can be considered as a part of circle and radius of that circle is called the radius of curvature of the path.

Consider that a particle is projected with velocity u = 10 m/s at an angle $\theta = 60^{o}$ with the horizontal and take value of g = 10 m/s^2: Now answer the following questions.

14. The radius of curvature of path of particle at the instant when the velocity vector of the particle becomes perpendicular to initial velocity vector is

(A) $\frac{20}{3\sqrt{3}}$m (B) $\frac{10}{3\sqrt{3}}$m

(C) $\frac{40}{3\sqrt{3}}$m (D) $\frac{80}{3\sqrt{3}}$m

15. The magnitude of acceleration of particle at that instant is

(A) 10 m/s^2 (B) $5\sqrt{3}\,m/s^2$

(C) 5 m/s^2 (D) $10\sqrt{3}\,m/s^2$

16. Tangential acceleration of particle at that instant is

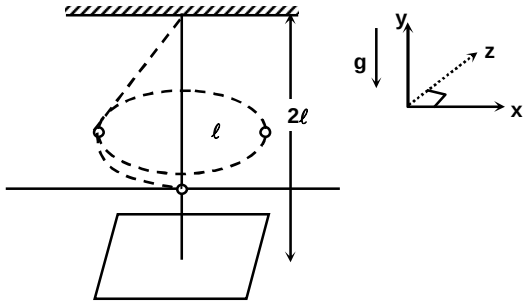
(A) 10 m/s^2 (B) 20 m/s^2

(C) 5 m/s^2 (D) $5\sqrt{3}\,m/s^2$

Space for rough work

$P_{17–19}$: Paragraph for Question Nos. 17 to 19

A pendulum bob has mass m. The length of pendulum is ℓ. It is initially at rest. A particle P of mass m/2 moving horizontally along –ve x-direction with velocity $\sqrt{2g\ell}$ collides with the bob and comes to rest. When the bob comes to rest momentarily, another particle Q of mass m moving horizontally along z-direction collides with the bob and sticks to it. It is observed that the bob now moves along a horizontal circle. The floor is a horizontal surface at a distance 2ℓ below the point of suspension of the pendulum.



17. Tension in string immediately after the first collision is
(A) 2mg
(B) mg
(C) $\frac{3}{2}$mg
(D) $\frac{5}{2}$mg

18. The height of circular path above the floor is
(A) $\frac{3\ell}{2}$
(B) $\frac{4\ell}{3}$
(C) $\frac{5\ell}{4}$
(D) data not sufficient

19. Time period of the circular motion
(A) $2\pi\sqrt{\frac{\ell}{g}}$
(B) $2\pi\sqrt{\frac{\sqrt{7}\ell}{4g}}$
(C) $2\pi\sqrt{\frac{\sqrt{7}\ell}{3g}}$
(D) $2\pi\sqrt{\frac{3\ell}{4g}}$

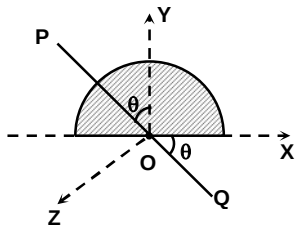
Space for rough work

SECTION – B

(Matching List Type)

This section contains **3 multiple choice questions.** Each question has matching Column(s). The codes for the Column(s) have choices (A), (B), (C) and (D) out of which **ONLY ONE or MORE** is/are correct

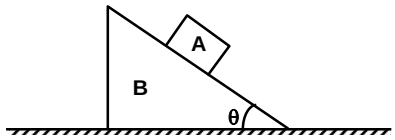
1. A uniform semi–circular disc of mass M and radius R lies in the X–Y plane with its centre at the origin as shown in figure. Also, axis PQ is in the X–Y plane. Then, match for the moment of inertia of semi–circular disc about different axis.



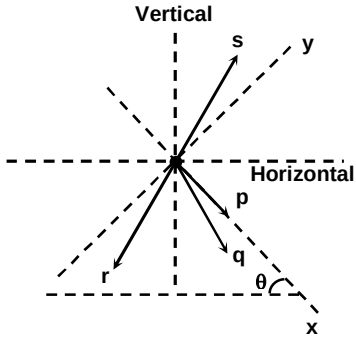
Column I		Column II	
(A)	I_X	(p)	$\frac{MR^2}{2}$
(B)	I_Y	(q)	$\frac{MR^2}{4}$
(C)	I_Z	(r)	$\frac{MR^2}{4}\cos^2\theta$
(D)	I_{PQ}	(s)	$\frac{MR^2}{2}\cos^2\theta$

Space for rough work

2. A block A is placed on wedge B, which is placed on horizontal surface. All the contact surfaces are rough but not sufficiently rough to prevent sliding at any surface. Match Column I and II. Column II indicates possible direction(s) X and Y axis are along incline and perpendicular to incline.

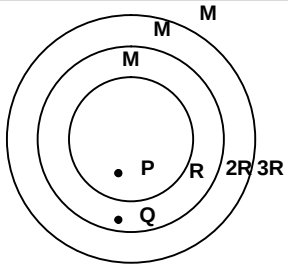
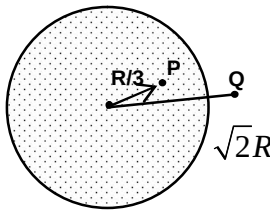
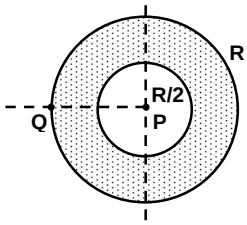
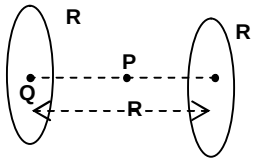


Column I		Column II	
(A)	Acceleration of A	(p)	
(B)	Net force applied by A on B	(q)	
(C)	Acceleration of A relative to B	(r)	
(D)	Net force applied by ground on B	(s)	



Space for rough work

3. E and V are the gravitational field and potential respectively.

Column-I		Column-II	
(A)	 Concentric charged spherical shells	(p)	$\frac{V_p}{V_q} < 1$
(B)	 solid uniformly charged sphere	(q)	$\frac{E_p}{E_q} < 1$
(C)	A charged spherical having inner radii R/2 and outer radii R. 	(r)	$\frac{V_p}{V_q} > 1$
(D)	 two identical co-axial charged rings	(s)	$\frac{E_p}{E_q} > 1$

Space for rough work

Chemistry

PART – II

SECTION – A

Straight Objective Type

This section contains **9 multiple choice questions** numbered 1 to 9. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct

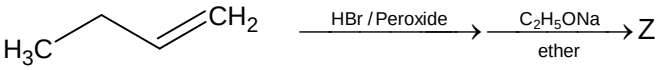
1. Cl_2 can displace I^- from KI but I displaces Cl^- from $KClO_3$. The statement is
   (A) True (B) Partly true
   (C) False (D) Can not be

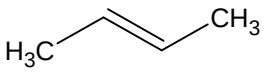
2. For a 4d electron the orbital angular momentum is
   (A) $\sqrt{6}\,\hbar$ (B) $\sqrt{12}\,\hbar$
   (C) $\sqrt{2}\,\hbar$ (D) zero

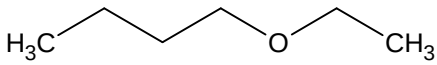
3. K_c for the reaction
   $A_{(g)} + B_{(g)} \rightleftharpoons C_{(g)} + D_{(g)}$, is 20 at 25°C. If a container contains 1, 2, 4, 5 moles per litre of $A_{(g)}$, $B_{(g)}$, $C_{(g)}$ and $D_{(g)}$ respectively at 25°C, then the reaction shall
   (A) proceed from left to right (B) proceed from right to left
   (C) be at equilibrium (D) None

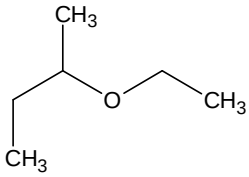
4. 10 ml of 0.2 M acid is added to 250 ml of a buffer solution with pH = 6.34 and the pH of the solution becomes 6.32. The buffer capacity of the solution is:
   (A) 0.1 (B) 0.2
   (C) 0.3 (D) 0.4

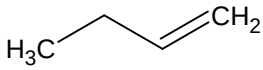
5. Identify Z in the following reaction sequence:



(A) 

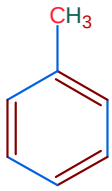
(B) 

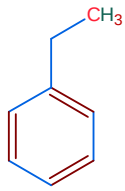
(C) 

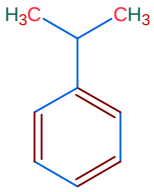
(D) 

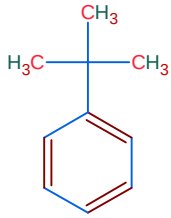
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6. Rate of electrophilic bromination in the aromatic nucleus will be maximum in

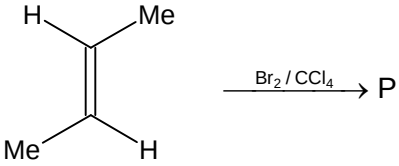
(A) 

(B) 

(C) 

(D) 

7. The incorrect statement about the product (P) of the reaction:



is which one of the following?

(A) P is optically inactive due to internal compensation
(B) P is optically inactive due to the presence of plane of symmetry in the molecule.
(C) The structure which P possesses can have four optical isomers possible.
(D) The structure of P can have three optical isomers possible.

8. For the gaseous reaction $A_{(g)} \longrightarrow 4B_{(g)} + 3C_{(g)}$ is found to be first order with respect to A. If at the starting the total pressure was 100 mm Hg and after 20 minutes it is found to be 400 mm Hg. The rate constant of the reaction is:

(A) $20\ min^{-1}$ (B) $1.2 \times 10^{3}\ sec^{-1}$

(C) $5.7 \times 10^{-4}\ sec^{-1}$ (D) $40\ min^{-1}$

9. The root mean square speed of an ideal gas at constant pressure varies with density (d) as

(A) d^2 (B) d

(C) $\frac{1}{\sqrt{d}}$ (D) $\sqrt{d}$

Space for rough work

Reasoning Type

This section contains **4 questions** numbered 10 to 13. Each question contains STATEMENT-1 (Assertion) and STATEMENT-2 (Reason). Each question has 4 choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct

10. STATEMENT-1: The alkali metals can form ionic hydrides which contain the H^- ion
and
STATEMENT-2: The alkali metals have low electronegativity.
(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
(C) Statement-1 is True, Statement -2 is False.
(D) Statement-1 is False, Statement-2 is True.

11. STATEMENT-1: Between $SiCl_4$ and CCl_4 only $SiCl_4$ reacts with water.
and
STATEMENT-2: Carbon does not contain empty d-orbitals.
(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
(C) Statement-1 is True, Statement -2 is False.
(D) Statement-1 is False, Statement-2 is True.

12. STATEMENT-1: Spectral line would not be seen for a $2p_x - 2p_z$ transition
and
STATEMENT-2: p–orbitals are degenerate orbitals
(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
(C) Statement-1 is True, Statement -2 is False.
(D) Statement-1 is False, Statement-2 is True.

13. STATEMENT-1: The heat absorbed during the isothermal expansion of an ideal gas against vacuum is zero.
and
STATEMENT-2: The volume occupied by the molecules of a real gas is negligible.
(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
(C) Statement-1 is True, Statement -2 is False.
(D) Statement-1 is False, Statement-2 is True.

Space for rough work

Comprehension Type

This section contains **2 groups of questions.** Each group has 3 multiple choice question based on a paragraph. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** is correct

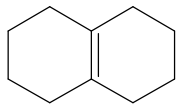
Paragraph for Question Nos. 14 to 16

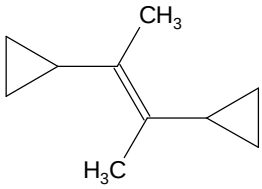
Read the following paragraph and answer the questions given below:

A hydrocarbon with molecular formula $C_{10}H_{16}$ undergoes reductive ozonolysis to give two moles of same compound having molecular formula C_5H_8O. Even the oxidative ozonolysis of the above hydrocarbon gives the same product. The hydrocarbon can react with three equivalents of H_2/Pd at 120^oC. Both the hydrocarbons and the ozonolysis products are not resolvable.
Choose the correct option:

14. The hydrocarbon is

(A) 

(B) 

(C) 

(D) Both (A) and (B)

15. Hydrocarbon shows
(A) geometrical isomerism
(B) position isomerism
(C) functional isomerism
(D) Tautomerism

16. Which of the following is correct?
(A) The ozonolysis product is optically active.
(B) The ozonolysis product is optically inactive racemic mixture.
(C) The ozonolysis product is optically inactive due to plane of symmetry.
(D) The ozonolysis product is optically inactive due to absence of chirality.

Space for rough work

Paragraph for Question Nos. 17 to 19

Read the following paragraph and answer the questions given below:

Stable equilibrium is of various types. Mechanical equilibrium is achieved when all particles are at rest and total potential energy of the system is minimum. At any stage where particles are at rest but the system is not at stable equilibrium as it can reduce its potential energy by reverting to another position, is called metastable equilibrium. Thermal equilibrium results from the absence of temperature gradients in the system. Chemical equilibrium is obtained when no further reaction occurs between reacting substances, i.e. forward and reverse rates of reaction are equal. When steam reacts with solid iron at high temperature, $Fe_3O_4(s)$ and hydrogen gas are produced. But the reaction never goes to completion. This is because as the products are formed, the reaction proceeds in reverse direction and when rate of reverse reaction is equal to rate of forward reaction, the concentration of reactants and products become constant and equilibrium is reached.

Choose the correct option:

17. The primary objective of the passage is
    (A) to discuss different types of equilibrium
    (B) to distinguish between different types of equilibrium
    (C) to discuss concept of mechanical and chemical equilibrium
    (D) to explain how changing state of equilibrium would influence internal energy

18. Which of the following is correct?
    (A) Chemical equilibrium can be stable, unstable or metastable.
    (B) The example given in the passage refers to thermodynamic equilibrium.
    (C) Addition of catalyst to a reaction in chemical equilibrium increases the amount of heat evolved in the reaction.
    (D) Addition of catalyst to a reaction in chemical equilibrium gives a new reaction path to the system.

19. The correct expression for equilibrium constant of the reaction, taken as an example of chemical equilibrium in the above passage is

    (A) $\frac{(pH_2)^2}{(pH_2O)^2}$ (B) $\frac{(pH_2)^4}{(pH_2O)^4}$

    (C) $\frac{(pH_2)^4[Fe_3O_4]}{(pH_2O)^4[Fe]}$ (D) $\frac{[Fe_3O_4]}{[Fe]}$

Space for rough work

SECTION – B

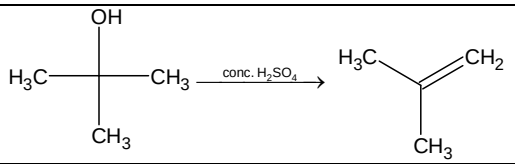
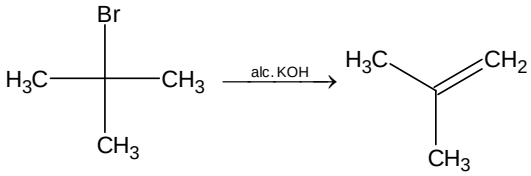
(Matching List Type)

This section contains **3 multiple choice questions.** Each question has matching Column(s). The codes for the Column(s) have choices (A), (B), (C) and (D) out of which **ONLY ONE or MORE** is/are correct

1. Match Column–I with Column–II and select the correct answer.

Column–I Ionic species		Column–II Shape	
(A)	SiF_5^-	(p)	Tetrahedral
(B)	ICl_4^-	(q)	Square planar
(C)	AsF_4^+	(r)	Linear
(D)	I_3^-	(s)	Trigonal bipyramidal

2. Match Column–I (reaction) with Column–II (reaction intermediate) and select the correct answer using the codes given below the lists:

Column–I		Column–II	
(A)	$CF_3 - CHCl_2 \xrightarrow{\text{alc. KOH}} CF_2 = CCl_2$	(p)	TS
(B)		(q)	Carbocation
(C)	$CH_3 - CH_2 - Br \xrightarrow{\text{alc. KOH}/\Delta} CH_2 = CH_2$	(r)	Carbanion
(D)		(s)	Free radical

3. Match the following:

Column - I (Number of Moles)		Column - II (Amount)	
(A)	0.1 mole	(p)	4480 mL of CO_2 at STP
(B)	0.2 mole	(q)	0.1 g atom of iron
(C)	0.25 mole	(r)	1.5×10^{23} molecules of O_2 gas
(D)	0.5 mole	(s)	9 mL water

Space for rough work

Mathematics PART – III

SECTION – A

Straight Objective Type

This section contains **9 multiple choice questions** numbered 1 to 9. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct

1. Vertices of a variable triangle are (3, 4), (5 cos θ, 5 sin θ) and (5 sin θ, 5 cos θ) then locus of its orthocentre is
(A) $(x + y - 1)^2 + (x - y - 7)^2 = 100$ (B) $(x + y - 7)^2 + (x - y - 1)^2 = 100$
(C) $(x + y - 7)^2 + (x + y - 1)^2 = 100$ (D) $(x + y - 7)^2 + (x - y + 1)^2 = 100$

2. The probability that at least one of the events A and B occur simultaneously with probability 3/5. If A and B occur simultaneously with probability 1/5 then $P(\bar{A}) + P(\bar{B})$ is
(A) $\frac{2}{5}$ (B) $\frac{4}{5}$
(C) $\frac{6}{5}$ (D) $\frac{7}{5}$

3. If a, b, c are three complex numbers selected randomly (without repetition) from the set {1, 2, 3…..2013) then the probability that $\arg\left(\frac{a-b}{a-c}\right) = 0$ is
(A) $\frac{1}{3}$ (B) $\frac{2}{3}$
(C) $\frac{3}{2013}$ (D) none of these

4. If one of the lines given by $6x^2 - xy + 4cy^2 = 0$ is $3x + 4y = 0$ then c is equal to
(A) –3 (B) –1
(C) 1 (D) 3

5. If the quadrate equation $4x^2 - 2(a + c - 1)x + ac - b = 0$ $(a > b > c)$
(A) both roots are greater than a
(B) both roots are less than c
(C) both roots be between $\left(\frac{c}{2}\right)$ and $\left(\frac{a}{2}\right)$
(D) exactly one of the roots has between $\left(\frac{c}{2}\right)$ and $\left(\frac{a}{2}\right)$

Space for rough work

6. If the focus of a parabola is at (0, –3) and its directrix is y = 3, then its equation is
(A) $x^2 = -12y$ (B) $x^2 = 12y$
(C) $y^2 = -12x$ (D) $y^2 = 12x$

7. Let z_k; k = 1, 2, 3, 4 be four complex numbers such that $|z_k| = \sqrt{k+1}$ and $|30z_1 + 20z_2 + 15z_3 + 12z_4| = \lambda|z_2z_3z_4 + z_3z_4z_1 + z_1z_2z_4 + z_1z_2z_3|$, then λ is equal to
(A) $|z_1\ z_2\ z_4|$ (B) $|z_2\ z_3\ z_4|$
(C) $|z_1\ z_3\ z_4|$ (D) $|z_1\ z_2\ z_3|$

8. A circle has the same centre as an ellipse and passes through the foci F_1 and F_2 of the ellipse the two curves intersect in four points. Let P be any point of intersection. If the major axis of the ellipse is 15 and the area of ΔPF_1F_2 is 26, then the distance between foci is
(A) 11 (B) 9
(C) 10 (D) 15

9. If normal chords at points P and Q on the parabola $y^2 = 4ax$ meets again on the parabola at R then the locus of the circum-centre of ΔPQR is
(A) $y^2 = 4a(x-a)$ (B) $2y^2 = a(x-a)$
(C) $2y^2 = a(x-2a)$ (D) none of these

Reasoning Type

This section contains **4 questions** numbered 10 to 13. Each question contains STATEMENT-1 (Assertion) and STATEMENT-2 (Reason). Each question has 4 choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct

10. STATEMENT 1: If P(2a, 0) be any point on the axis of parabola, then the chord QPR, satisfy $\frac{1}{PQ^2} + \frac{1}{PR^2} = \frac{1}{4a^2}$

STATEMENT 2: There exists a point on the axis of the parabola $y^2 = 4ax$, such that $\frac{1}{PQ^2} + \frac{1}{PR^2}$ is equal to constant for all chord QPR of the parabola
(A) Both the statements are true and Statement 2 is correct explanation of Statement 1
(B) Both the Statements are true and Statement 2 is not the correct explanation of Statement 1
(C) Statement 1 is true and Statement 2 is false
(D) Statement 1 is false and Statement 2 is true

Space for rough work

11. STATEMENT 1: 3 is a multiple root of order 2 of the equation $x^3 - 5x^2 + 3x + 9 = 0$
STATEMENT 2: If $f(x) = x^3 - 5x^2 + 3x + 9$, then $f''(3) = 0$
(A) Both the statements are true and Statement 2 is correct explanation of Statement 1
(B) Both the Statements are true and Statement 2 is not the correct explanation of Statement 1
(C) Statement 1 is true and Statement 2 is false
(D) Statement 1 is false and Statement 2 is true

12. STATEMENT 1: The function $f(x) = (3x - 1)|4x^2 - 12x + 5| \cos \pi x$ is differentiable at $x = \frac{1}{2}, \frac{5}{2}$

STATEMENT 2: $\cos(2n+1)\frac{\pi}{2} = 0, \forall n \in I$
(A) Both the statements are true and Statement 2 is correct explanation of Statement 1
(B) Both the Statements are true and Statement 2 is not the correct explanation of Statement 1
(C) Statement 1 is true and Statement 2 is false
(D) Statement 1 is false and Statement 2 is true

13. If positive numbers x, y, z satisfy $x + y + z = 1$, $(x, y, z < 1/2)$ then
STATEMENT 1: $\left[\frac{(1-2x)+(1-2y)+(1-2z)}{3}\right] \geq (1-2x)(1-2y)(1-2z)^{1/3}$
STATEMENT 2: For any three positive numbers a, b, c their AM $\geq$ GM
(A) Both the statements are true and Statement 2 is correct explanation of Statement 1
(B) Both the Statements are true and Statement 2 is not the correct explanation of Statement 1
(C) Statement 1 is true and Statement 2 is false
(D) Statement 1 is false and Statement 2 is true

Space for rough work

Comprehension Type

This section contains **2 groups of questions.** Each group has 3 multiple choice question based on a paragraph. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** is correct

Paragraph for Question Nos. 14 to 16

Read the following write up carefully and answer the following questions:
Inequalities are very important to understand in mathematics.
$a > b$, $a < b$, $a \geq b$, $a \leq b$ and $a \neq b$ are called inequality terms. Many times calculus help us to prove inequalities e.g. To check which is greater of the two from e^{π} or π^{e} we take $f(x) = x^{1/x}$ and check, where the function increases and so on. Now answer the following questions

14. If $a = \pi^{3e}$, $b = 3^{\pi e}$, $c = e^{3\pi}$, then
(A) $a < b < c$ (B) $b < a < c$
(C) $c < a < b$ (D) $a < c < b$

15. The largest interval for which $x^{12} - x^9 + x^4 - x + 3 - \sin \pi x + \cos \pi x > 0$ is
(A) $(-1, 1)$ (B) $(-\infty, 2)$
(C) $(-\log 12, \log 12)$ (D) $(-\infty, \infty)$

16. If $f(x)$ is a decreasing function and $f(0) = 0$ then
(A) $f(3) > 0$ (B) $f(2) < 0$
(C) $f(-3) < 0$ (D) none of these

Space for rough work

Paragraph for Question Nos. 17 to 19

Read the following write up carefully and answer the following questions:

$(1+x)^n = c_0 + c_1x + c_2x^2....c_nx^n$

17. Total number of term, that are dependent of the value of x, in the expansion of $\left(x^2 - 2 + \frac{1}{x^2}\right)^n$ is equal to

(A) 2n + 1 (B) 2n
(C) n (D) n + 1

18. In the expansion of $\left(a^3 + \frac{1}{a^3} + 1\right)^{100}$ number of distinct terms is

(A) 200 (B) 201
(C) 202 (D) none of these

19. An equation $a_0 + a_1x + a_2x^2 + ...a_{99}x^{99} + x^{100} = 0$ has roots ${}^{99}C_0, {}^{99}C_1, ...{}^{99}C_{99}$. Find the value of a_{98} is

(A) $2^{198} - {}^{198}C_{99}$ (B) $\frac{2^{198} + {}^{198}C_{99}}{2}$

(C) $2^{99} - {}^{99}C_{49}$ (D) none of these

SECTION – B

(Matching List Type)

This section contains **3 multiple choice questions.** Each question has matching Column(s). The codes for the Column(s) have choices (A), (B), (C) and (D) out of which **ONLY ONE or MORE** is/are correct

1. Match the following Column–I with Column–II

	Column – I		Column – II
(A)	If the normal at one end of a latus rectum of an ellipse of eccentricity e passes through one end of minor axis, then $e^2 + e^4$ is equal to	(p)	$\frac{\sqrt{5}}{2}$
(B)	The chords of contact of the pair of tangents to the circle $x^2 + y^2 = 1$ drawn from any point on the line $2x + y = 4$ pass through the point $\left(\frac{1}{\lambda}, \frac{1}{2\lambda}\right)$, then λ is equal to	(q)	123
(C)	The eccentricity of the hyperbola $x^2 - 4y^2 = 1$ is	(r)	1
(D)	The number of lines passing through (3, 4) at a distance 5 units from (6, 8) are	(s)	2

Space for rough work

2. Let A and B are two independent events such that $P(A) = \frac{1}{3}$ and $P(B) = \frac{1}{4}$. Now match the following Column–I with Column–II

Column – I		Column – II	
(A)	P(A∪B) is equal to	(p)	$\frac{1}{12}$
(B)	P(A∪B′) is equal to	(q)	$\frac{1}{2}$
(C)	P(B∪A′) is equal to	(r)	$\frac{5}{6}$
(D)	P(A∩B) is equal to	(s)	$\frac{3}{4}$

3. Match the following Column–I with Column–II

Column – I		Column – II	
(A)	x dy = y(dx + y dy), y > 0 y(1) = 1 and $y(x_0) = -3$, then $x_0 = \lambda$, then λ is less than	(p)	$\frac{1}{4}$
(B)	If y(t) is solution of $(t+1)\frac{dy}{dt} - ty = 1$, y(0) = –1, and y(1) = λ, then λ is greater than	(q)	–15
(C)	$(x^2 + y^2)dx = xy\,dx$ and y(1) = 1 and $y(x_0) = \sqrt{e}$, then $x_0 = \lambda$, then λ is greater than	(r)	$-\frac{1}{2}$
(D)	$\frac{dy}{dx} + \frac{2y}{x} = 0$, y(1) = 1, then y(2) is equal to	(s)	$\sqrt{3}e$

Space for rough work